Example Simulation

Introduction to Simulation

Data has been collected at a service centre to enable the analysis and redesign of the service system.

Customers arrivals are described by the distribution

Inter-arrival Times	8	12	16	20
Frequency percentage	20	40	20	20

Service times are given by

Service Duration	7	11	15	18	20	22	24
Frequency percentage	30	20	10	15	15	5	5

Simulate the arrival of 30 customers, assuming that only the quoted times could occur, what does your simulation indicate about this system?

Customer	**Inter-arrival Time (IAT)**	**Arrival Time (AT)**	**Service Starts (SS)**	**Service Duration (SD)**	**Service Ends (SE)**	**Customer Waiting Time**	**Service Waiting Time**
1	Generate using random number	IAT(1)	AT(1)	Generate using random number	SS(1) + SD(1)	SS(1) – AT(1)	SS(1)
2	Generate using random number	IAT(1) + IAT(2)	Max{SE(1), AT(2)}	Generate using random number	SS(2) + SD(2)	SS(2) – AT(2)	SS(2) - SE(1)

Investigate the effect of changing the last inter-arrival value. E.g.

Inter-arrival Times	8	12	16	20
Frequency percentage	20	40	20	20

Inter-arrival Times	8	12	16	30
Frequency percentage	20	40	20	20

Sensitivity: Internal